

5.6.5.3	Creation of Action Plan.....	73
5.6.5.4	Implement Action Plan	73
5.6.5.5	Monitor, Evaluate, Revise.....	73
5.7	Workforce Planning.....	73
5.8	The Future of Workforce Planning is Data-Driven.....	74
5.8.1	Curating the Learning Process: Competencies May Be Built and Evaluated.....	74
5.8.2	Making the Experience for Skill Activation: Activating and Progressing.....	74
5.8.3	Transactable, Transferable, and Objective Skills.....	74
5.9	Impact of Skills Framework on Business Outcomes.....	75
5.10	Data-Driven Team Building	75
5.10.1	Data-Driven Work Culture	75
5.10.2	Quality Data for Better Decision-Making.....	76
5.10.3	Data Literacy	76
5.10.4	Data Is an Asset.....	76
5.10.5	Data as an Integral Part of the Organization.....	76
5.11	Principles for Success of Data-Driven Decision-Making.....	76
5.11.1	The Three Important Facts of Workforce Planning: “What, Where, and Why” (3Ws).....	77
5.11.2	Recognize Data Gaps and Quality Problems	77
5.11.3	Specify Ownership and Roles	77
5.11.4	Visualize the Data’s Underlying Meaning.....	77
5.11.5	Focus on the Target Audience	78
5.11.6	Encourage Iteration.....	78
5.11.7	Utilization and Adoption of Measures.....	78
5.11.8	Locate Patterns	79
5.12	Recommendations	79
5.13	Conclusion	81
	References.....	82

5.1 INTRODUCTION

It’s hard to overstate the need for people with data science acumen and some data-specific skills. The best talents will go to the best tech-oriented companies, so it’s hard for small and mid-size companies to recruit and retain the best and the brightest.

– **Mark Daniel Ward, a Professor at Purdue and Director of the Data Mine (The Data Mine, 2022).**

The nature of work changes as a result of solid disruption forces such as demographic changes, shifting career paths, and unrelenting technological advancements. A shift in skill sets that can adapt to shifting work environments and demographics, as well as potential industry-level disruptions brought on by the widespread adoption of technology like artificial intelligence (AI), deep learning,

and automation, are the driving forces behind these changes (Babasaheb, Sphurti & Khang, 2023).

Organizations compete with one another and, with time, keep up with the rising rate of change. Automation (Hajimahmud et al., 2022), AI, and other technological marvels can only take businesses so far. The success of any corporation still depends on its human personnel. Their inventiveness, creativity, empathy, and other unmistakably human traits hold the key to creating the breakthroughs and possibilities businesses will require to prosper in the future.

However, today's workforce still has a way to go before it is fully future-ready. New talents and competencies must be created and fostered, and work itself must be redefined. The key to this transition will be learning while doing (Khang et al., 2023).

In particular, the abilities that will be more crucial over the next five to ten years will almost entirely fall into the following category of twenty-first-century skills. Twenty-first-century competencies is a catch-all phrase that refers to a wide range of skill sets and competencies regarded as crucial in the twenty first century.

Data-driven decision-making (DDDM, 2018): In general, is using facts, measurements, and data to inform strategic business decisions that align with your goals, objectives, and initiatives.

Importance of DDDM: Businesses may produce real-time insights and predictions to improve their performance using data-based decision-making. In this way, companies can evaluate the efficacy of various methods and decide which business actions will result in sustained growth (Snehal, Babasaheb & Khang, 2023).

5.2 DATA-DRIVEN DECISION-MAKING AND SUCCESS

5.2.1 INCREASED ACCOUNTABILITY AND TRANSPARENCY

This involves being transparent as to which employee earns what particular salary and why, taking into account factors like a person's tenure at the organization. It prevents people from being paid unfairly and brings transparency and accountability.

5.2.2 INNOVATION AND STEADY IMPROVEMENT

Data-based decision management inadvertently creates room for innovation and development. Organizations can make additional modifications and monitor essential indicators based on collected data. For example, a business may be pushed in the proper direction by customer feedback.

5.2.3 QUICK DECISION-MAKING

Precise analytical insights aid in the resolution of numerous business issues. Entrepreneurs frequently need to mine data to gain prescriptive and predictive insights when using data-based decision management.

For example, deciding on production facilities is one of the most common examples of decision-making in management. You will have to expand your production capacity and choose how much capacity installation is necessary to satisfy demand.

Along with that, you will also need to determine the workforce required to operate the production processes as well as the appropriate equipment for the job (Babasaheb, Sphurti, & Khang, 2023).

5.2.4 ADDITIONAL REMARKS ON MARKET RESEARCH

Organizations can develop new goods, services, and workplace activities and discover trends using DDDM. By looking at historical data, companies can predict what to expect in the near future and what they should adjust to improve performance and compete. For example, they can create questionnaires and survey forms to collect data and further predict customer needs.

5.2.5 EXCEPTIONAL ADAPTABILITY

Last but not least, DDDM has the significant advantage of making your company very adaptive.

The digital world is constantly evolving, and to keep up with the constantly shifting environment around you, you must use data to inform and empower data-driven business decisions. You may, for example, expand and change your business over time by embracing digital data, which will make your company more flexible.

5.3 FRAMEWORK FOR DATA-DRIVEN WORKFORCE IDENTIFICATION

For the success of any industry, the main contribution is from the workforce. If an organization wants to be future-ready, identification of the future skill requirements for the workforce is essential. The data is generated at a tremendous rate, and utilization of the data for skill identification is profitable for the industry. With the help of data, informed decisions can be taken to plan the set of actions to train the workforce.

Informed decision-making helps organizations to achieve lifetime success, which may seem unpredictable at the moment. DDDM is a formula for winning consistently. In the traditional form of leadership, it was difficult to track what exactly the winning factors were and repeat them in the future, but with the data-driven framework, it has become smooth and easy.

With effective data utilization, organizations achieve assurance and confidence in understanding the present and future requirements, which can be easily managed by training the existing staff.

The data-driven workforce identification is a robust framework for bridging the gaps and identifying the expertise of the existing employee pool and future requirements. The proposed framework for data-driven workforce identification is shown in [Figure 5.1](#).

The business problem is identified or reiterated initially to begin data collection for future skill requirements. Once the data is available based on the upcoming assignment, the skill and workforce requirements are highlighted. The workforce skill requirements are known based on the data collected; accordingly, the training and recruitment of the skilled workforce can be planned to build the future workforce.

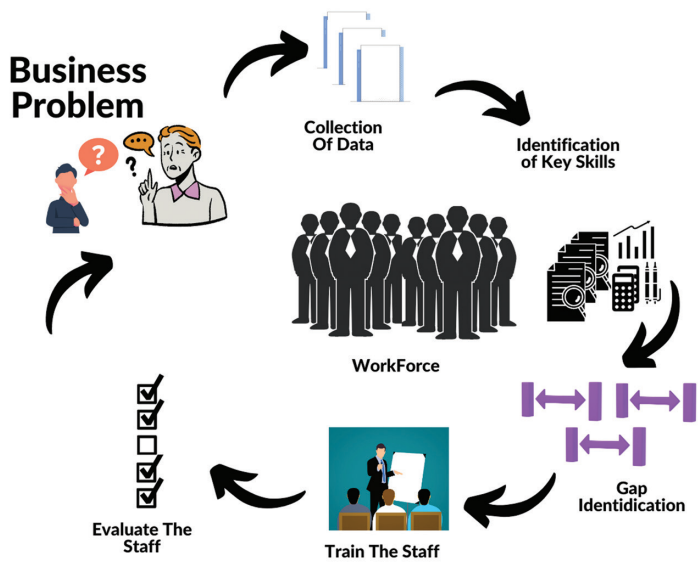


FIGURE 5.1 Proposed framework for data-driven workforce identification.

Before the pandemic, the talent hunt was directly based on targeting the market-place. But with an enormous change in the employment market due to the pandemic, the focus has now shifted towards foresight the future workforce requirements and opportunities and optimally utilizing the full potential of existing employees by planning and prioritizing the growth of the organization and the employees by training them.

In this scenario, data-driven workforce planning is a decisive and intellectual path for bridging the gap between the required skills and existing workforce knowledge. With the help of data in workforce planning, organizations gain affirmation and trust in understanding the future and current workforce requirements so they can be effectively managed by utilizing the existing players.

5.4 BENEFITS OF USING DATA FOR WORKFORCE PLANNING

Data is a “New Oil (2022)”¹; it gives a wide range of information to excel in the business by identifying previously unseen gaps. It saves a significant amount of time and helps organizations to prioritize training and development. The data-driven workforce identification helps in various ways.

5.4.1 WORKFORCE SCHEDULING

The daily data of workforce operations can help organizations identify permanent and temporary demand for skilled resources and optimal utilization of the resources. The data helps identify challenges to gain the required output for achieving the

business objectives. It will help the organization to plan and schedule the resources so that no project will be deprived of them. It allows organizations to make optimal utilization of the resources (Tailor, Pareek & Khang, 2022).

5.4.2 SKILL GAP IDENTIFICATION

With the help of data about the performance of the employees at the individual and organizational levels, information is collected about more high-performing employees so that more mentoring responsibilities can be given to them. Organizations can identify the noticeable gaps and train the workforce to make them future-ready.

5.4.3 EMPLOYEE EXPERIENCE AND SATISFACTION

The data collected about employee experience and satisfaction can be crucial to the leadership in planning the next set of activities. It is observed that happy employees contribute more to the organization's development. With this, top performers and budding newcomers can be rewarded and recognized for their contributions. It deepens the sense of mutual trust and respect among the employees. It reinforces enhanced loyalty, pushes the performances to peak level, increases the longevity of the performers, and attracts the prospective best new employees.

5.4.4 EMPLOYEE GRIEVANCE REDRESSAL

One of the significant aspects of keeping employee morale and trust high involves anticipating the team's needs. The data analysis can tell when there is an employee shortage and alert the leadership team about it before it becomes an emergency. This type of situation increases the employees' burnout. With the proper study of employee demands, workforce planning can be executed without any stress to the existing workforce.

5.4.5 UTILIZATION OF AUTOMATIZATION

Automatization is a compelling application of AI that can generate data reports to get critical information, helping the leadership prioritize crucial matters such as the workforce required for the next quarter, achieving goals, and customers by updating the details to maintain long-term relationships (Khang, Rani & Sivaraman, 2022).

5.5 TRADITIONAL AND DATA-DRIVEN WORKFORCE PLANNING

Putting traditional and data-driven workforce planning into practice may achieve several goals. It can assist you in determining which sectors in your company might benefit by adding groups and those where your workers would profit from further training. However, you can do a few essential things by effectively putting workforce planning strategies into effect.

Problem avoidance is one aim of good staff planning. Making a comprehensive evaluation of the skills each department needs and determining whether its

employees have them is a crucial component of workforce planning. By performing this evaluation, you'll be able to spot areas where abilities are lacking and be in a position to address the problem before it causes difficulties.

5.5.1 WORKFORCE PLANNING USING DATA-DRIVEN SUCCESS FOR ORGANIZATIONS

To comprehend how the present situation and external influences are affecting the business, HR directors require proof. The planning process, a crucial aspect of workforce planning, enables the workforce planning teams to foresee the impact of outside influences. Leaders may use the findings to anticipate different future paths and workforce needs.

5.5.2 WORKFORCE PLANNING COMPRISES OF THREE MAIN APPROACHES

These three approaches are strategic, operational, and tactical. Each has a distinct objective and planning level. There is no one-size-fits-all method to workforce planning.

5.5.3 INDIVIDUALS' DATA FOR OPTIMIZED WORKFORCE PLANNING

HR analytics or people analytics may also collect, track, and analyze worker data to enhance corporate goals and pave the way for new proposals. CFOs can see talent shortfalls, stress, or excessive turnover rates using performance management, tenure, and payroll data.

HR executives and financial experts may use data to evaluate the state of the workforce and develop a strategy that boosts the sense of achievement and enhances retention. With the capacity to dig down from summary to detail and produce ad hoc models, HR analytics provide valuable tools for fine-tuning and managing the strategy (Khang et al., 2022).

5.5.4 THREE FORMS OF HUMAN RESOURCE ANALYTICS

5.5.4.1 Historical Analytics

Historical analytics consider the past to measure and characterize workforce changes. They provide information on revenue per FTE, employee satisfaction, pay scales, and the average time to fill open positions, benefit expenses, and head count trends.

5.5.4.2 Predictive Analytics

The use of statistics and modeling approaches to forecast future results and performance is known as predictive analytics. With predictive analytics, data trends in the past and present are examined to see if they are likely to recur. They enable companies and investors to change where they allocate their resources to profit from potential future occurrences. To determine skill and personnel demands, future company goals are considered.

5.5.4.3 Prescribed Analytics

Prescribed analytics blend historical and predictive analytics to reshape the workplace and assessment of results. Based on your short- and long-term company strategies, you may use data analytics to establish if you have the proper number of employees with the relevant talents in each area at any given moment. Leaders in workforce planning may make choices about the workforce more quickly and confidently by visualizing and modeling HR situations.

5.6 DATA USAGE FOR WORKFORCE PLANNING

Data emit a broad beam of light that illuminates previously hidden areas of company procedures and reveals holes that had not previously been seen. They help prioritize focus and efforts for changes and saves time. To best make use of the strength and energy of analytics, most businesses find it helpful to concentrate on four main areas.

5.6.1 EFFECTIVE STAFF PLANNING

To estimate both steady and changing need levels, identify the best resources to handle the tasks and difficulties put forward, and describe the output level necessary for finishing the project, the data that reflects what is happening in everyday activities must be collected and analyzed. This ensures that no initiative is overlooked or deprived of necessary resources to accomplish its objectives. It guarantees that no essential resources are ever wasted or overused at crucial times.

5.6.2 IDENTIFY SPOT TALENT GAPS

It is possible to validate whether players are excellent performers (and can subsequently coach team members) and to determine the general requirements for every specialized department by being able to examine quality standards both individually and in groups. By matching the task with workers who can expand their skill base the most readily, this technique makes it easy to fill any apparent gaps.

5.6.3 NURTURING EMPLOYEE SATISFACTION

When bright entrants and top performers routinely receive the praise and accolades they merit for their contributions to the company's success, a culture of mutual trust and respect develops, growing over time. The prospect of establishing longevity among top performers and luring the most significant potential entrants is made possible by data-driven worker insights that further build loyalty and boost peak quality standards.

5.6.4 RESOLVING EMPLOYEE CONCERNS

Anticipating the job requirements of the entire team is another approach to maintaining employee morale and corporate trust at high levels. For instance, frequent

and continuing data analysis enables the early detection and effective management of staff shortages, which is when worker burnout becomes notably higher.

However, with efficient and appropriate oversight of staff demands, all easily constructed and maintained with workforce planning, the staff continues to work in an optimal situation, one that is conducive to working at top performance levels and being inspired by the successful outcomes they consistently achieve.

5.6.5 IMPLEMENTATION OF DATA-DRIVEN WORKFORCE STRATEGY

5.6.5.1 Establish a Strategic Direction

The first stage is to educate the workers on the plans and initiatives required to meet short- and long-term goals and objectives.

5.6.5.2 Examine, Identify, and Carry Out

Fully appreciating workforce assets, comprehending fluctuating capabilities through time and also between periods, designing and preparing for evolving employee requirements, and being conscious of gaps between the present and future skilled workforce are essential to the success of any data-driven workforce strategy.

5.6.5.3 Creation of Action Plan

Create an action plan, which entails coming up with a viable plan to fill in the holes that have been discovered, putting that plan into action, and then keeping an eye on the results to gauge progress and direction.

5.6.5.4 Implement Action Plan

At this level, necessary cooperation, messaging, and marketing efforts are in flux. Financial and human resources are already in place, and responsibilities are specified.

5.6.5.5 Monitor, Evaluate, Revise

Continual monitoring, reporting on milestone accomplishments and progress, evaluating improvement opportunities, and making modifications to stay on course are all part of regular management.

5.7 WORKFORCE PLANNING

Workforce planning is an essential element for any organization's success. It ensures that the required number of employees with relevant skills are available within the organization at the appropriate time to achieve desired organizational goals. While planning the workforce, the organization needs to examine three things.

First, chief data officers should identify the correct number of workers required based on data that should be available to organizations to achieve business objectives.

Second, customers need real-time and on-the-spot services as in the current era. So businesses need to plan the resource requirements well and make them accessible as and when required.

Third, organizations must maintain the right talent for the skilled workforce to meet current and future needs.

5.8 THE FUTURE OF WORKFORCE PLANNING IS DATA-DRIVEN

Skills must be developed in tandem if the company is seeking agility and adaptive responsiveness to answer changing needs. It will enable long-lasting action and outcomes. A change from competencies to skills is necessary for this. Four essential elements often make up competencies:

- **Knowledge:** Learnable, quantifiable, and objective.
- **Skills:** Acquirable, quantifiable, and objective.
- **Attributes:** Due to the substantial subjectivity and prejudice ingrained in observations, attributes talk about “who you are” and seldom change.
- **Behaviors:** Hard-to-learn behaviors that can be changed, but doing so will take more focus and discipline. Managers typically assign subjective values to behaviors.

On the other hand, the data-based skills enhancement approach is centered on the following three principles.

5.8.1 CURATING THE LEARNING PROCESS: COMPETENCIES MAY BE BUILT AND EVALUATED

Organizations must establish the appropriate development focus for employees to perform successfully in their existing jobs and advance their work. The intended result is to learn more. Because the emphasis on skills enables workers to serve better and develop their careers, they study more at work.

5.8.2 MAKING THE EXPERIENCE FOR SKILL ACTIVATION: ACTIVATING AND PROGRESSING

Skills is simple to learn anytime and anywhere. Therefore, employers must provide workers the chance to practice, make mistakes, recover, and develop their abilities while at work. Because workers put their abilities to use in actual work situations, the goal of this stage is to help them grow more quickly.

5.8.3 TRANSACTABLE, TRANSFERABLE, AND OBJECTIVE SKILLS

These are necessary to provide the analytics-insights experience. This emphasis will offer insights and knowledge about talent supply, skill movement, and skill velocity to influence business results. Showing results is the intended result of this step. Dynamic skill data makes it possible to see training efforts’ consequences and improve business decisions (Bhambri et al., 2022).

The organizational learning plans must include all three of these events. To do this, Learning and Development (L&D) must create an environment that fosters learning through stretching assignments, mentoring, etc. These fully immersive experiences will aid workers in developing their confidence in their recently acquired abilities and learning new ones. Since this is an ongoing process, continued leadership commitment is necessary.

5.9 IMPACT OF SKILLS FRAMEWORK ON BUSINESS OUTCOMES

The following four steps must be taken to create a skills-enablement journey:

- Development via knowledge acquisition and assessment: L&D must increase employee learning engagement to increase learning development and delivery effectiveness.
- By incorporating learning experiences into the regular course of work, L&D may increase the efficacy of learning. This is known as skills activation. It will hasten the rate at which talents change.
- Human resources must make it possible for workers to exchange skills and expertise. This is done through talent planning and workforce deployment. Programs must be created to support professional advancement. Identifying talent readiness for workforce deployment and proactively planning for talent migration based on strategic needs are good places to start. It will assist in lowering reliance on contract employees.
- Workforce planning, organizational design, and enabling performance contribute to organizational success and sustainability. Success in using skills requires democratizing a data-driven approach to high-potential (HiPo) programs identification and development and enhancing succession planning and leadership effectiveness. To foster an inclusive and high-performing culture that will promote organizational success, HR and business executives must work together.

5.10 DATA-DRIVEN TEAM BUILDING

Data is power, and power always comes with responsibility. The question is how the teams utilize the data to build a better future for organizations and employees. The essential/critical point to be realized is the data can help the company to achieve market objectives. However, the importance of data still needs to get its due credit in many organizations. The data-driven teams can achieve success faster based on insight and making the right decisions about upskilling and future resource requirements.

5.10.1 DATA-DRIVEN WORK CULTURE

According to various research, organizations adopting and actively engaging in data-driven decisions in their daily work planning are more successful.

The data play a crucial role in creating a common platform to align the employees to the business goal and motivate them to achieve the required objectives. It helps the employees to visualize how their contribution is helping the organization to follow the right path toward achieving the business goal. This increases the accountability and visibility of individuals within the organization.

Data-driven culture promotes collaboration among the team members and outside the team. It breaks the traditional style of working in silos and increases information sharing. It also increases transparency across the business, which can help set the strategies to achieve the desired goals.

5.10.2 QUALITY DATA FOR BETTER DECISION-MAKING

According to Prof. Dr. Alex Khang who is a data analysis expert, it is very important to look after the quality factors of the data in this study, such as accuracy, relevance, reliability of the source, and completeness. Therefore, accessibility of data of all the employees for decision-making or strategic planning of the next set of actions. Quality data is needed to build trust in data-driven decision-making based on that data set (Khang et al., December 29, 2022).

5.10.3 DATA LITERACY

In recent generations, computer literacy is mandatory. In time, all organizations will recruit employees with data literacy as a mandatory skill. According to our research, organizations using data as the prime element in strategic planning are 23 times more likely to acquire customers, six times more likely to retain customers, and 19 times more likely to be profitable.

Considering all these facts, many organizations are making data literacy an essential success factor and training employees to have trust and confidence in data usage for decision-making. Empowering employees with data literacy teaches them to make the best decisions using data.

5.10.4 DATA IS AN ASSET

Organizations should practice keeping data at the center of the business and present the data as an essential asset. It is crucial for the management, employees, and all stakeholders to understand the importance of quality data and how to use it in better decision-making. Organizations should enforce the data at various levels of the organization to ensure data-driven culture. Treat data as an integral part of the organization.

5.10.5 DATA AS AN INTEGRAL PART OF THE ORGANIZATION

Define the business goals using data. Keep the key performance indicators (KPIs) and other organization performance metrics transparent using data-driven strategies. Re-evaluate the previous performance and strategy using data. Set expectations that data will be used to evaluate performance and decisions. This will increase accountability for decisions and ensure data always back them.

5.11 PRINCIPLES FOR SUCCESS OF DATA-DRIVEN DECISION-MAKING

For decision-making to be objective and well-informed, data and analytics are crucial. When the proper procedures are followed, businesses may use data to improve customer service, streamline their supply chains, and gauge the success of their marketing campaigns. A few principles of DDDM help companies use the data correctly (ADDDM, 2022).

5.11.1 THE THREE IMPORTANT FACTS OF WORKFORCE PLANNING: “WHAT, WHERE, AND WHY” (3Ws)

Organizations frequently gather a lot of data without thinking about why they need that data, its use, and where and how they will use it. It leaves them with a surplus of confusing data. With this comes our first principle: to answer the question of what, where, and why?

For instance, I want to gather information on students who fail to receive the minimum passing grade in one of the subjects, to determine whether a particular student is struggling to comprehend the material or if the professor is lacking clarity. For this, we first establish the 3Ws.

What: Take some time upfront to determine what solutions you’re seeking – e.g., to decide whether to start with a few remedial classes or to replace the professor.

Why: Explain why you are seeking those solutions and what you plan to do with them once you have them. For example, we are looking for answers to these questions to determine the caliber of instructors at the university and then take appropriate action.

Where: Ascertain the source of the solutions. For example, we can accomplish this by gathering student opinions of teachers’ effectiveness.

5.11.2 RECOGNIZE DATA GAPS AND QUALITY PROBLEMS

Data-driven firms try to explain gaps or quality issues when they are found. This may involve revisiting your systems and imposing stricter data entry standards, or it may entail developing a new approach to collect the desired data.

The gap between operating perception and customer expectation of managers at a supermarket, for instance, might believe that customers desire more brand-name products and self-checkout counters. Still, the customers are more concerned with the cleanliness and friendliness of the staff.

5.11.3 SPECIFY OWNERSHIP AND ROLES

Because the roles and ownership are not clearly defined, businesses make decisions where infrastructure is not controlled, and projects are never approved. Before launching their data projects, data-driven companies decide who will be responsible for what and establish the roles and duties.

Therefore, once the “why” and “what” data are needed for are clarified, we move on to where the solution is coming from and then move on to who will handle what data and how they will handle it. It will help firms to be organized and also help in faster and more effective working and execution – e.g., assigning specific departmental roles, assigning leaders to each department, etc.

5.11.4 VISUALIZE THE DATA’S UNDERLYING MEANING

An essential step in the data analysis process is data visualization. A dashboard is effective when users can quickly and easily find the necessary information. A table of

numbers is essentially meaningless without context. You'll be able to spot trends and draw inferences about the data more quickly by making exciting charts and graphs.

For example, think as a customer and designer as well as an analyst. Dashboards should follow a precise, logical arrangement, contain interactive components that are easy to find, and have a straightforward design that makes complex selections easier.

5.11.5 FOCUS ON THE TARGET AUDIENCE

Ensure your analytics are done so your target audience will want to know more about the data. The audience is left wanting more when analytics are done well.

For instance, provide further insights. Good analytics begin with providing answers to inquiries. In addition to asking more specific questions, going into greater detail about a particular topic and gaining insights into brand-new subject areas is essential.

More probing queries: You should start looking for relationships between the different subject areas rather than monitoring and measuring them separately. The conversation evolves from "What happened?" to "What IF this happens?" This leads to further inquiries that wander into the field of data science.

Optimized Data: This entails attempting to close the data gaps that were initially noted. Data optimization refers to collecting and managing data efficiently to maximize the speed and effectiveness of extracting, analyzing, and utilizing critical information.

5.11.6 ENCOURAGE ITERATION

Organizations that rely on data are aware that data analytics are iterative. This is a process that can and ought to be repeated. The idea that everything should be constructed and perfected in one go makes the task feel overburdened. Spend time preparing the data, layering in the data comprehension, reviewing the outcomes, and executing your solution. Start over or repeat the cycle a few times before finalizing something.

For example, Coke has experimented with many versions of its well-known Coca-Cola beverage, including the unsuccessful New Coke variant. The goal is to provide the best-tasting product, which consumers will enjoy.

5.11.7 UTILIZATION AND ADOPTION OF MEASURES

Understanding user behaviors, from consumption to creation, requires measurement. The three recommended subject areas to focus on are user engagement, utilization, and performance. For example,

- **User Engagement:** How often and how long users interact with your website or application is measured by user engagement. This indicator reveals whether your audience values the offers you make, whether they be goods or services. Companies monitor user engagements, such as clicks, downloads, and shares, to measure engagement.

- **Utilization:** A KPI called resource utilization assesses work output and effort over a given time window (or capacity). Project managers can anticipate resource availability across many categories when resources are used to their full potential.
- **Performance:** The process of gathering, evaluating, and reporting data about a person, group, organization, system, or component's performance is known as performance measurement.

5.11.8 LOCATE PATTERNS

Finding a pattern or a correlation between various data pieces is at the core of data analysis. Insights and inferences can be derived from these patterns and relationships. For example, we often collect data to find patterns in the data, like numbers trending upward or correlations between two sets of numbers.

5.12 RECOMMENDATIONS

Future skills are crucial for building functional abilities and enhancing your professional prospects (Workforce of the Future, 2022). Companies need these skills. Thus, regardless of a candidate's career path, employers may hire people committed to honing these skills. Future skills should be those that can help you in the following ways:

- Becoming more flexible.
- Adapting to new environments.
- Adapting to recent career changes.
- Achieving success in a variety of contexts.
- Offering greater benefits to companies so that they hire you.

This section discusses the recommendation presented by Deloitte (2021) on how organizations can upskill their employees to achieve the required objective of getting a skilled workforce and making them future-ready.

- A sustainable workforce can meet today's challenges and adapt to tomorrow's challenges. This is the future-ready workforce that is designed to be flexible, adaptable, and resilient. For example, organizations can use analytics to test scenarios and inform decisions at a lower level by reducing work to the task level. It helps them better control what gets done, by whom or what, and where. It enables them to stop performing tasks that are no longer required and delegate the remaining work—the essential, critical tasks—to the ideal combination of permanent staff, contract workers, and automation.
- Nowadays, a workforce that is constantly learning in the flow of work. The culture of lifelong learning is ingrained and incorporated into the work process in the era of digital economy. It is a far cry from the traditional, classroom-based learning we have been using for so long. It is learning that is immediately applicable and available in real time, as and when needed – for example, creating a highly effective learning environment by integrating

learning into tasks that are already a part of employees' regular tasks and using resources they are already familiar with. This is fundamentally different from on-the-job training.

- A workforce built on enduring capabilities.
 - A focus on building and developing enduring capabilities first and necessary skills second is another characteristic of the future-ready workforce. It consists of imagination, empathy, curiosity, resilience, creativity, emotional intelligence, teaming, social intelligence, sense making, critical thinking, and adaptive thinking.
 - For instance, many tasks humans perform can be done by machines, though humans still have an advantage in some situations. People are better at relating to and comprehending other people's needs, recognizing and adjusting to shifting settings, and coming up with original, imaginative new solutions. In the years to come, organizations will be well-positioned to acquire a strategic edge if they embrace, develop, and nurture enduring human talents.
 - Their workforces will be able to recognize and adapt to change and quickly pick up the skills required at that precise time to keep moving forward and prosper in the climate of constant disruption that is quickly becoming the norm.
- Workforce planning and analytics should be used to rethink and redesign the work broadly. Understanding the internal and external elements that will impact an organization's business and employees in the coming years is the first step in workforce planning and analytics (Rana et al., 2021).
 - For example, consider a business that now offers two distinct product lines and is considering adding a third. Sales of the first product line are gradually falling while those of the second are accelerating and are expected to rise.
 - Leaders can decide on a revenue target for the future year using information such as present revenue, predicted annual growth, and the number of sales and support workers. It is a streamlined method of determining how staff should develop in light of these goals and the current information.
 - It is safe to assume that the first product line is less profitable if staffing expenses are the same for both product lines. Therefore, there is probably no need to hire more employees for the first product line. If there are "extra" people, they are sent to the second product line, presuming their abilities are still applicable there (if not, the business could consider retraining them). In addition, managers can prepare and retrain staff in advance of demand for the second product line, lowering hiring expenses.
 - According to officials' calculations, the second product line will continue to be more profitable. The company can then use the growth rate to calculate the approximate number of new employees needed to handle the increase. Depending on how these projections turn out, these hires may occur quickly or gradually.

5.13 CONCLUSION

In conclusion, using data in workforce planning is essential for the success of any industry. By identifying future skill requirements and utilizing data to make informed decisions, organizations can effectively train their workforce and ensure they are prepared for the future (Morris et al., 2023).

DDDM (2018) is a proven formula for achieving success. By embracing this approach, organizations can gain confidence and assurance in their ability to understand and manage their workforce requirements (Khanh & Khang, 2021).

The ongoing COVID-19 pandemic has highlighted the need for a shift in focus from targeting the marketplace to forecasting future workforce requirements and opportunities, and data-driven workforce planning is a crucial tool in this effort. However, the importance of data in achieving these goals is often undervalued in many organizations, and teams must recognize the power and responsibility that comes with utilizing data effectively to build a better future for their organizations and employees.

Organizations compete with one another and, with time, keep up with the rising rate of change. Automation, AI, and other technological wonders can only take businesses so far; technology is just a part of the answer. The success of every corporation still depends on its human personnel.

You can complete the trip by following data-driven transformational concepts. It takes a lot of work to use information effectively so that your company can become data-driven. Your ability to quickly ingest, integrate, and analyze your data—current internal data and the expanding flood of big data from new internal and external sources—and transform it into insights you can act upon at the point of action is essential (Rani et al., 2023).

Future-ready skills can help you solve issues and constructively deal with them. Technological changes affect each industry differently, and these abilities can help you adjust to the changing workforce and working environment. Acquiring these innovative abilities can raise your profile and position you as a future asset for a business or employer. However, today's workforce still has a way to go before it is fully future-ready.

New talents and competencies must be created and fostered, and work itself must be redefined. The key to this transition will be learning while doing. There is now a strategy that high-performance firms may use to create and maintain high-quality workforce planning initiatives and remove the long-standing obstacles to efficient workforce planning.

The organization must promote a culture of data-driven planning and be prepared to place equal weight on the planning process and the plan itself.

The managers taking part in this planning process will therefore be more prepared to make choices as company outcomes and predictions change and become more adept at modeling how business objectives and conditions change and necessitate various skill sets. The company must also invest in workforce planning software that provides employee-level action planning, bottom-up demand planning, predictive supply analytics, and gap summarizing.

A high-performance organization can undertake gap analyses on the work unit, business unit, and organization as a whole by using this method. Managers will be

more capable of anticipating and adapting to changes in the business at the work unit level.

The relevance of a job or a person's effectiveness will allow managers to make data-driven decisions that eventually transition from general human resources policies to tailored human resources interventions. The personnel planning program will help both business divisions and the entire company (Misra et al., 2023).

REFERENCES

- DDDM. (2018). A Guide to Data-Driven Decision Making: What It Is, Its Importance, & How to Implement It. Tableau. Retrieved December 13, 2022, from <https://www.tableau.com/learn/articles/datadriven-decision-making>
- ADDDM. (June 20, 2022). Advantages of Data-Driven Decision Making. Softjourn. Retrieved December 13, 2022, from <https://softjourn.com/insights/data-driven-decision-making>
- Babasaheb, J., Sphurti, B., & Khang, A., "Design of Competency Models in the Human Capital Management System," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches*, (1st ed.) (2023), pp. 32–50. CRC Press. <https://doi.org/10.1201/9781003357070-3>
- Babasaheb, J., Sphurti, B., & Khang, A., "Industry Revolution 4.0: Workforce Competency Models and Designs," *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches*, (1st ed.) (2023), pp. 14–31. CRC Press. <https://doi.org/10.1201/9781003357070-2>
- Bhambri, P., Rani, S., Gupta, G., & Khang, A., *Cloud and Fog Computing Platforms for Internet of Things* (2022). CRC Press. <https://doi.org/10.1201/9781003213888>
- Deloitte. (2021). Building the future-ready workforce - Unleash the potential of your organization and people. *Deloitte*. Retrieved December 13, 2022, from <https://www2.deloitte.com/content/dam/Deloitte/ca/Documents/consulting/ca-future-ready-workforce-en-aoda.pdf>
- Hajimahmud, V.A., Khang, A., Hahanov, V., Litvinova, E., Chumachenko, S., & Alyar, A.V., "Autonomous Robots for Smart City: Closer to Augmented Humanity," *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) (2022). CRC Press. <https://doi.org/10.1201/9781003252542-7>
- Khang, A., Chowdhury, S., & Sharma, S. (Eds.). *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.) (2022). CRC Press. <https://doi.org/10.1201/9781003269281>
- Khang, A., Hahanov, V., Litvinova, E., Chumachenko, S., Hajimahmud, V.A., & Alyar, A.V., "The Key Assistant of Smart City – Sensors and Tools," *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) (2022). CRC Press. <https://doi.org/10.1201/9781003252542-17>
- Khang, A., Rani, S., Gujrati, R., Uygun, H., & Gupta, S.K., (Eds.). *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023). CRC Press. <https://doi.org/10.1201/99781003357070>
- Khang, A., Rani, S., & Sivaraman, A.K., *AI-Centric Smart City Ecosystems: Technologies, Design and Implementation* (1st ed.) (2022). CRC Press. <https://doi.org/10.1201/9781003252542>
- Khanh, H.H., & Khang, A., "The Role of Artificial Intelligence in Blockchain Applications," *Reinventing Manufacturing and Business Processes Through Artificial Intelligence*, (2021), pp. 20–40. CRC Press. <https://doi.org/10.1201/9781003145011-2>
- Misra, A., Shah, V., Khang, A., & Gupta, S.K. (Eds.). *AI-Aided IoT Technologies and Applications in the Smart Business and Production* (1st ed.) (2023). CRC Press. <https://doi.org/10.1201/9781003392224>

- Morris, G., Babasaheb, J., Khang, A., Gupta, S.K., & Hajimahmud, V.A., *AI-Centric Modelling and Analytics: Concepts, Designs, Technologies, and Applications* (1st ed.) (2023). CRC Press. <https://doi.org/10.1201/9781003400110>
- New Oil. (2022). Data as the New Oil Is Not Enough: Four Principles for Avoiding Data Fires. Retrieved from <https://www.forbes.com/sites/nishatalagala/2022/03/02/data-as-the-new-oil-is-not-enough-four-principles-for-avoiding-data-fires/>
- Rana, G., Khang, A., Sharma, R., Goel, A.K., & Dubey, A.K. (Eds.). *Reinventing Manufacturing and Business Processes Through Artificial Intelligence* (2021). CRC Press. <https://doi.org/10.1201/9781003145011>
- Rani, S., Bhambri, P., Kataria, A., Khang, A., & Sivaraman, A.K. (Eds.). *Big Data, Cloud Computing and IoT: Tools and Applications* (1st ed.) (2003). Chapman and Hall/CRC. <https://doi.org/10.1201/9781003298335>
- Snehal, M., Babasaheb, J., & Khang, A., “Workforce Management System: Concepts, Definitions, Principles, and Implementation,” *Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches* (1st ed.) (2023), pp. 1–13. CRC Press. <https://doi.org/10.1201/9781003357070-1>
- Tailor, R.K., Pareek, R., & Khang, A. (Eds.). “Robot Process Automation in Blockchain,” *The Data-Driven Blockchain Ecosystem: Fundamentals, Applications, and Emerging Technologies* (1st ed.) (2002), pp. 149–164. CRC Press. <https://doi.org/10.1201/9781003269281-8>
- The Data Mine. (2022). <https://datamine.purdue.edu/>
- Workforce of the Future. (2022). 5 Ways Employers Can Drive Skills Development for the Workforce of the Future. *Coursera Blog*. Retrieved December 13, 2022, from <https://blog.coursera.org/5-ways-employers-can-drive-skills-development-for-the-workforce-of-the-future/guides>



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6 Data Mining Processes and Decision-Making Models in the Personnel Management System

*Shashi Kant Gupta, Alex Khang, Parin Somani,
Chandra Kumar Dixit, and Anchal Pathak*

CONTENTS

6.1	Introduction	86
6.2	Problem Statement.....	88
6.3	Proposed Methodology.....	89
6.3.1	Data Collection.....	90
6.3.2	Preprocessing.....	90
6.3.2.1	Data Sampling.....	91
6.3.2.2	Data Cleaning	91
6.3.2.3	Data Integration	91
6.3.3	Feature Extraction Using Principle Component Analysis.....	91
6.3.4	Genetic Algorithm Based Feature Selection	93
6.3.5	Wavelet Fuzzy Logic-Based Deep Neural Network (WFL-DNN).....	94
6.3.5.1	Deep Neural Network	94
6.3.5.2	Fuzzy Logic	94
6.3.5.3	Wavelet Transformation	96
6.3.5.4	Wavelet Fuzzy Logic-Based Deep Neural Network	97
6.4	Performance Analytics	98
6.4.1	Accuracy	99
6.4.2	Precision	100
6.4.3	Recall	100
6.4.4	Decision-Making Skill	100
6.4.5	Execution Time.....	100
6.5	Conclusion	102
	References.....	102

6.1 INTRODUCTION

An increasing number of companies are using open innovation as a means of discovering the finest personnel for their organizations and managing the situation in an ever-changing environment.

Scholars and practitioners alike are increasingly concerned about employee engagement as a management problem. Given the substantial data linking employee attitudes, behavior, performance, and well-being to a company's financial and customer performance indicators, this is hardly a surprise (Grade & Ghadimi, 2021).

It's widely accepted that human capital is one of the most valuable assets a business may have in the knowledge economy to get a competitive advantage. A huge amount of money is invested because companies and major institutions care about human capital (Armutat, 2018).

As a result, many organizations place a high value on the process of hiring new employees, seeing it as a key turning point in the organization's growth. This helps in the acquisition of the entire industry in which the company operates, as well as determining the quality of the company's human resources and the people it hires (Müller-Camen et al., 2019).

Recruitment is the process of finding and interviewing potential workers to see whether they meet the company's needs in terms of qualifications and experience (Pak et al., 2019).

Organizational success depends on having a sound personnel management system in place, which helps the company achieve strategic objectives while also producing high levels of performance indicators.

It is critical for the success of the enterprise to build a creative, competent team that takes into consideration the unique features of each employee. Forming an environment that is both morally and psychologically supportive is essential as well. One of the most important indications of a leader's performance is the status of the organization's sociopsychological environment and the extent to which it affects each member (Macke & Genari, 2019).

When forming work groups and teams, giving power, and allocating responsibilities, employers must consider the personalities of their workers. Creating a work environment that is both pleasant and productive for each employee is essential (Carnevale & Hatak, 2020).

An efficient human management system must use the most advanced and cutting-edge technology available (Mousa & Othman, 2020).

The following blocks represent the types of personnel management systems that are most receptive to new ideas (Snehal, Babasaheb & Khang, 2023).

- Personnel planning subsystem.
- Personnel development subsystem.
- The subsystem of stimulation and motivation of personnel.
- The subsystem of organization of innovative activity of personnel.

Each company's top executives recognize that the efficiency of their operations is directly tied to the expertise of their workforce (Khang et al., 2023).

It is not always possible to fix an issue via internal training inside an organization, particularly when a certain skill set is required at the time. Personnel selection in the real world is difficult, since decision-makers prefer to make decisions and forecasts based on qualitative approaches, such as interviewing the applicants and getting to know them via talks and group activities (Amrutha & Geetha, 2020).

According to their quantitative measurements, such as productivity, outputs of their contributions, etc., they may have a skewed view of the team and individual success. To invest in human capital, then need to be explicit about what they are doing and avoid using ambiguous terminology (Chams & García-Blandón, 2019).

A neutrosophic theory is often used in decision issues to move this forward. As a result of the neutrosophic study, it has been possible to pick suppliers, design supplier selection criteria, select intelligent medical equipment, and evaluate supply chain hazards (Ruparel et al., 2020) as shown in [Figure 6.1](#) (Votto et al., 2021).

Solving difficult issues and constructing interrelationships and interdependencies among criteria and options are both possible uses for a neutrosophic set. As per Protalinsky, Khanova and Shcherbatov (2019), an HR manager’s responsibilities

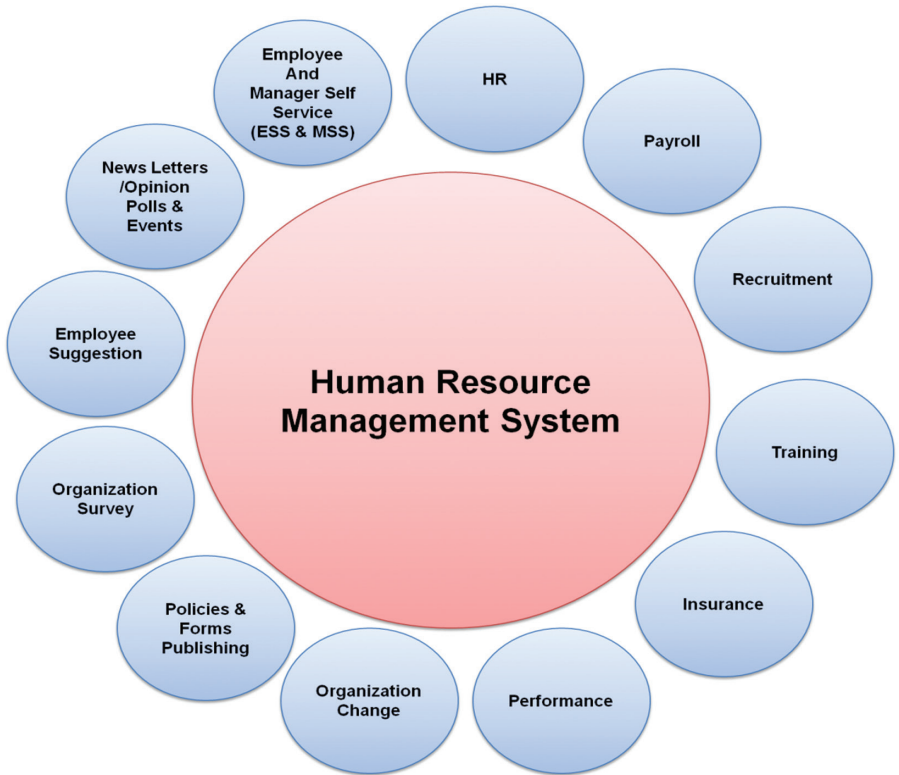


FIGURE 6.1 A human resource management (HRM) system.

encompass anything from also before the planning to strategic recruitment to staff training, to growth pay management, to worker relations, to health care, and to service providing. Improvement of organizational effectiveness, participative management, and the quality of work are all part of this plan.

As a result, employee voices had a higher influence on the association between personal management and emotional weariness and engagement when employee voices were low. This is partly because personal management programs in the public sector tend to be top-down and concentrate more on measurement, monitoring, and control than on development (Bhoi, Pujari & Balabantaray, 2020; Shah et al., 2024).

As a result, personnel management was seen as more of a need than a resource in this research (Sheikh, Tiwari & Singhal, 2019).

Using data mining techniques, human resources management (HRM) may benefit from discovering new information (Zhang, Brown & Li, 2019). By applying a wavelet fuzzy logic-based deep neural network (WFL-DNN), this study has established a framework that helps human resources monitor the performance of their employees and make precise judgments so that the efficacy of the HRM system may be improved.

6.2 PROBLEM STATEMENT

One of the most crucial aspects of a business is its human resources. To be successful, a business must place a high value on the management of its human resources. Sadly, many firms fail to recognize the value of human resources and instead rely on outmoded techniques of personnel management, which may hurt the company's productivity (Babasaheb, Sphurti & Khang, 2023a).

It is well-known that perseverance in attaining objectives is only established in workers when strong motivational elements are present. A lack of motivation means that a person's labor is done just to escape punishment rather than to get a reward.

A new set of motivating and disciplinary procedures must be established in the firm to execute new qualitative and quantitative positive improvements in its growth. That's why it's critical to have a well-oiled system of employee incentives in place (Khang, Rani & Sivaraman, 2022).

A real-world solution is an open-ended question. Because of this, many optimization stages are needed to handle the data. Fuzzy logic was used to analyze and process the information to lessen the level of uncertainty.

The survey investigated some of the most fundamental logic to identify the issue and find a solution (Hajimahmud et al., 2022).

However, when used to simulate more complicated systems, it has several limitations. Machine learning algorithms and ID3 (Iterative Dichotomiser 3) aren't good enough for complicated datasets (In decision tree learning, ID3 is an algorithm invented by Ross Quinlan used to generate a decision tree from a dataset. ID3 is the precursor to the C4.5 algorithm and is typically used in the machine learning and natural language processing domains).

In this chapter, the performance of an employee is being assessed, and hence the dataset is unique for each individual (Rana et al., 2021). This model's high accuracy

and execution time were both compared to several other models to see where we might make improvements.

6.3 PROPOSED METHODOLOGY

As a business grows, so does its human resources (HR) department, which is responsible for a variety of tasks, including recruiting, training, and managing employees. It is the progress of each employee’s job and the monitoring of his or her objectives that determine the performance of an employee (Babasaheb, Sphurti & Khang, 2023b).

The HR employee is one of those responsible for ensuring that each employee’s evaluation is fair and objective. An automated data management unit with staff metrics is needed to ensure that these operations are conducted truthfully. The schematic representation of the suggested optimization is depicted in Figure 6.2.

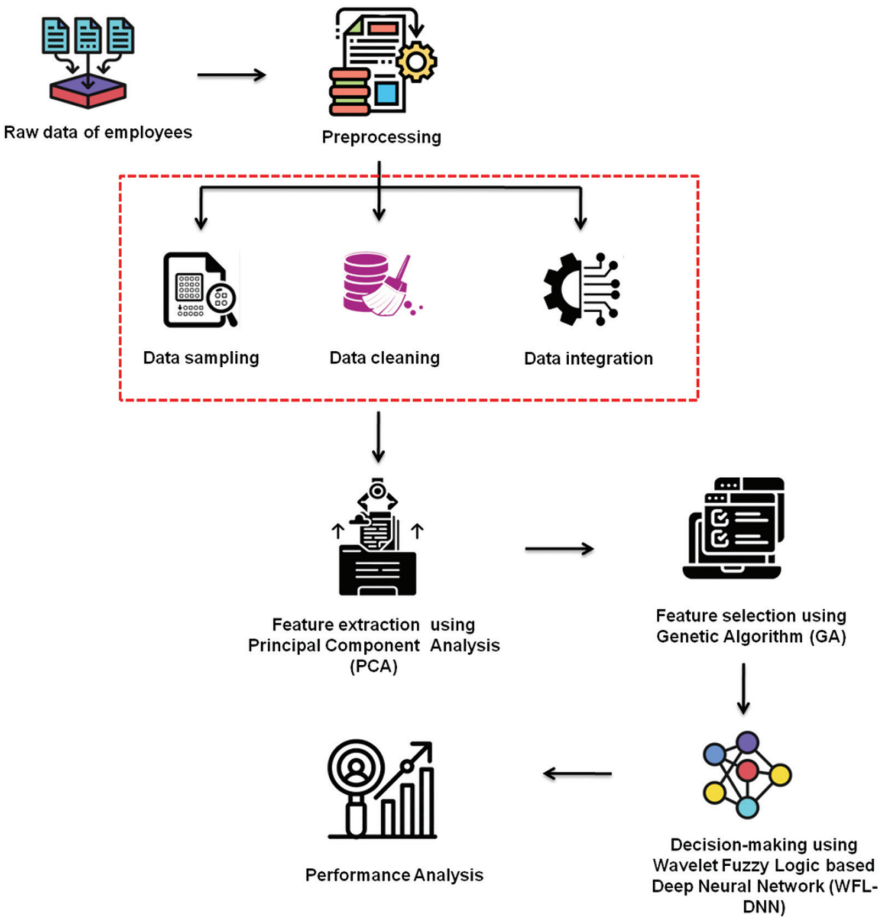


FIGURE 6.2 Schematic representation of the suggested optimization.

TABLE 6.1
Representation of attributes

S. No	Attributes	Description
1.	AR	Awards Received
2.	TM	Teaching Methodology
3.	JP	Journal Publication
4.	HO	Honors Obtained
5.	RP	Research Progress
6.	CP	Conference Presented
7.	EXE	External Examiner
8.	QPA	Question Paper Setting
9.	ODF	Organizing Department functions
10.	UVE	University Evaluation

6.3.1 DATA COLLECTION

Selecting the data to be taken into account is done using the individually updated list. [Table 6.1](#) shows the qualities of 10 workers, as demonstrated in the sample. [Table 6.1](#) lists the values of each of the 10 characteristics that were mentioned for each employee.

Numbers or Boolean responses are available for each attribute. Boolean and numerical responses may be either true or false, as shown in [Table 6.2](#).

6.3.2 PREPROCESSING

In data mining, preprocessing is one of the most important responsibilities, since it involves getting the data ready for the mining process. It's a goal of data preprocessing to shrink datasets, discover relationships among them, standardize them, eliminate outliers, and extract information from them, among other things. Data cleansing and integration are among the approaches included, as shown in [Figure 6.3](#).

TABLE 6.2
Dataset Description

Employee ID	AR	TM	JP	HO	RP	CP	EXE	QPA	ODF	UVE
ID 1 (16059)	0.0	Yes	2.0	0.0	No	0.0	Yes	Yes	No	No
ID 2 (16060)	0.0	Yes	0.0	0.0	No	0.0	Yes	No	No	No
ID 3 (16061)	0.0	Yes	2.0	0.0	No	2.0	Yes	Yes	No	No
ID 4 (16062)	1.0	No	1.0	0.0	No	1.0	Yes	No	Yes	Yes
ID 5 (16063)	0.0	Yes	1.0	0.0	No	0.0	Yes	No	Yes	Yes
ID 6 (16064)	0.0	Yes	0.0	1.0	No	1.0	No	Yes	Yes	Yes
ID 7 (16065)	1.0	Yes	1.0	1.0	Yes	1.0	Yes	Yes	Yes	Yes
ID 8 (16066)	1.0	No	0.0	1.0	No	0.0	No	Yes	No	No
ID 9 (16067)	0.0	Yes	0.0	0.0	No	0.0	No	Yes	No	Yes
ID 10 (16068)	0.0	Yes	0.0	1.0	No	0.0	No	No	No	No

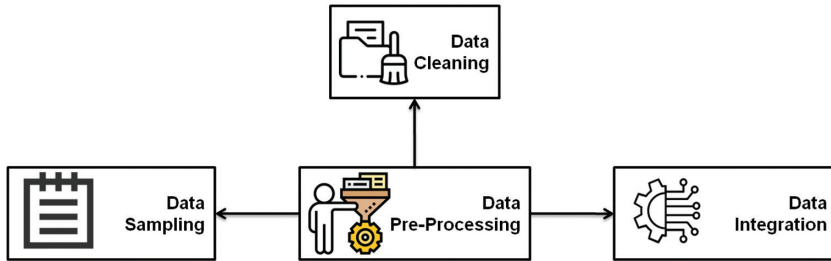


FIGURE 6.3 Four steps in the data preprocessing lifecycle.

In order to discover and complete null values, repair incoherence, and standardize outliers to decrease noise, data cleaning is necessary.

There are several ways to clean data, but the first step is to merge data sets from storage resources like servers and databases. The datasets are then rebuilt to fill in missing values and de-duplicated to normalize the data (Bhambri et al., 2022).

Verification and enrichment of the data is the following phase, which entails verifying the quality of the data and adding further changes so that it may be analyzed and decided upon (Khanh & Khang, 2021).

6.3.2.1 Data Sampling

Sampling, a data preprocessing strategy, may be used to successfully address the issue of data imbalance aggregation. Problems of unequal data distribution may be addressed with this technique (Rani et al., 2023).

6.3.2.2 Data Cleaning

Unresolved issues with row data might include missing records, noise values, outliers, and inconsistent data. The first stage in data preparation procedures is data cleaning, which is used to discover missing values, revise noisy data (noisy data is meaningless data), recognize outliers, and fix inconsistencies.

The mining process will be adversely affected by the messy data, resulting in unreliable and subpar output. As a result, the usage of data cleansing methods is critical.

6.3.2.3 Data Integration

Data integration is the process of bringing together data from several sources into a single database. When integrating data, it is important to address issues such as data redundancy, inconsistency, duplication, and the like. It is a preprocessing approach in data mining that merges data from many heterogeneous datasets to keep and support an overall view of the data. Data warehousing merges data from several sources into one cohesive database.

6.3.3 FEATURE EXTRACTION USING PRINCIPLE COMPONENT ANALYSIS

Principal component analysis (PCA) can be viewed as a collection of orthogonal linear transformations of the original variables, with the goal of the dynamic process

being to keep as much information contained in the original factors as practically possible from the perspective of reducing the number of dimensions.

Let R be an $n \times s$ data matrix, where n and s represent the number of factors and observations, accordingly. Assume the entire Y column means are zero for the sake of presenting convenience. $U_1 = \sum_y' = \alpha_1 y R_y$, where $\alpha_1 = (\alpha_{11}, \dots, \alpha_{1s})^Q$, Q , is the definition of the first principal component. Q is selected to maximize U_1 's variance, i.e.,

$$\alpha_1 = \arg \max_{\alpha} \alpha^Q \sum \alpha \text{ subject to } \alpha_1 = 1 \quad (6.1)$$

with $\Sigma = (R^Q R)/n$. Sequential definitions for the remaining primary components are as follows:

$$\alpha_{g+1} = \arg \max_{\alpha} \alpha^Q \sum \alpha \quad (6.2)$$

depending on

$$\|\alpha\| = 1 \text{ and } \alpha_h = 0, \forall 1 \leq h \leq g \quad (6.3)$$

According to this definition, the first g eigenvectors are the first g loading vectors.

The singular value decomposition (SVD) of R is connected to PCA by its eigen-decomposition formulation. Assume R denotes SVD.

$$R = ZLG^Q \quad (6.4)$$

where U and G are orthonormal matrices of $n \times s$ and $s \times s$ rows and columns, respectively, and D is a diagonal matrix with diagonal components $s_1, \dots, s_t$ in descending order. G is the loading matrix of the main components because the columns of G are the eigenvectors.

We can see that $U_g = Z_g d_g$ since $RG = ZL$, Z_g is the g^{th} column of Z . Recognize that the SVD is a good low-rank estimate of the data matrix.

An alternative geometric interpretation of PCA yields a linear manifold as the closest approximation to the observed data. This concept aligns with how PCA is built. Make ix the x^{th} row in Y .

Take the first g main components together, which equals $K_g = [K_1 | \dots | K_g]$. K_g is a $s \times g$ orthonormal matrix by definition. Each observation should be projected to the linear region covered by $\{K_1, \dots, K_g\}$.

The projected data are $T_g R_x$, $1 \leq x \leq m$ and the projection operator is $T_g = K_g K_g^Q$. By reducing the overall h_2 approximation error, one may determine the optimal projection.

$$\min_{B_g} \sum_{x=1}^m \|i_x - B_g B_g^Q i_x\|^2 \quad (6.5)$$

The illustration of the solution's primary first-generation components is simple. A wide range of scales and unit types may be used for application parameters. The

marginal variance of each variable is set to 1 as a result of standardization for each parameter.

With this approach for principal component analysis (PCA), we will get the correlation matrix of raw data and the resulting covariance matrix of standardized variables after you are done. Remember that the eigenvalues of the correlation matrix may vary from the eigenvalues of the covariance.

6.3.4 GENETIC ALGORITHM BASED FEATURE SELECTION

In the field of machine learning, genetic algorithms (GAs) are an important subset of simulated annealing (SA). Based on the theory of biological evolution, GAs are optimization algorithms. A GA has a chromosomal group where proposed solutions to these issues may be found. There are unresolved portions on every chromosome.

A random distribution of 1s and 0s makes up the bulk of the chromosomal population. Between 0 and 1, a variety of ranges are presented. In this encoding system, the length of chromatics is determined by the number of primary components in the region at the top.

Chromosomes provide clues to possible solutions or a collection of critical elements once again for a potential competitor. The quest for GAs aids in the growth of the species as shown in Figure 6.4.

Random function-dependent probability is used to construct the overall fitness function in the evolutionary process defined in the equation:

$$Fitness(J\backslash a) = sum\ xin\ aJa(x)/J*(Ja(x)) \tag{6.6}$$

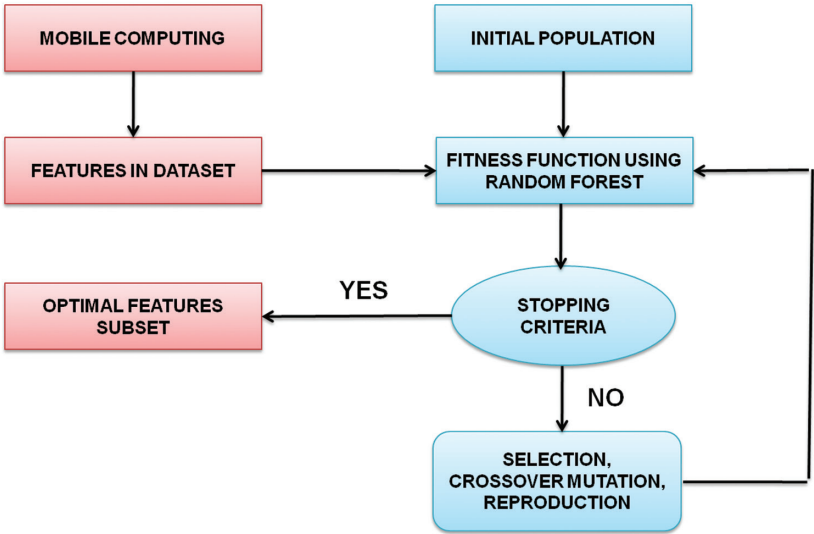


FIGURE 6.4 Genetic algorithm workflow.

where, $Ja(x)/J$ = The number of things within the data source, a = the property of J , $J^*(Ja(x))$ = Team probability of the items with the property a .

6.3.5 WAVELET FUZZY LOGIC-BASED DEEP NEURAL NETWORK (WFL-DNN)

6.3.5.1 Deep Neural Network

A deep neural network (DNN) is a collection of neurons that are arranged in layers (Rani et al., 2022). Each layer gets information from the neurons of the layer that comes before it, and each layer is responsible for a certain function. The neurons in the network provide a sophisticated and non-linear mapping from the input to the output (Tailor, Pareek & Khang, 2022).

To learn this translation from the input, a loss feed-forward algorithm is used to make adjustments to the weights of each neuron in the network. A representation of examples of neural networks and deep neural networks are shown in Figures 6.5 and 6.6.

An illustration of the notion of an artificial neural network is shown as a neuron in the top half as Figure 6.6. In addition to being conceptual, the mechanism by which the DNN generates content is easily understood by most people (Khang et al., 2023).

6.3.5.2 Fuzzy Logic

Fuzzy sets were the inspiration for fuzzy logic, which was developed later. In a fuzzy set, each component is given a degree of membership, which is often represented by a number in the range of 0 to 1.

Developing fuzzy logic requires attributing varying degrees of validity to the various arguments. Real numbers are those that fall in the range [0, 1], where 00 represents “totally false,” 11 represents “entirely true,” and the other digits represent “partial truth,” often known as transitional degrees of reality.

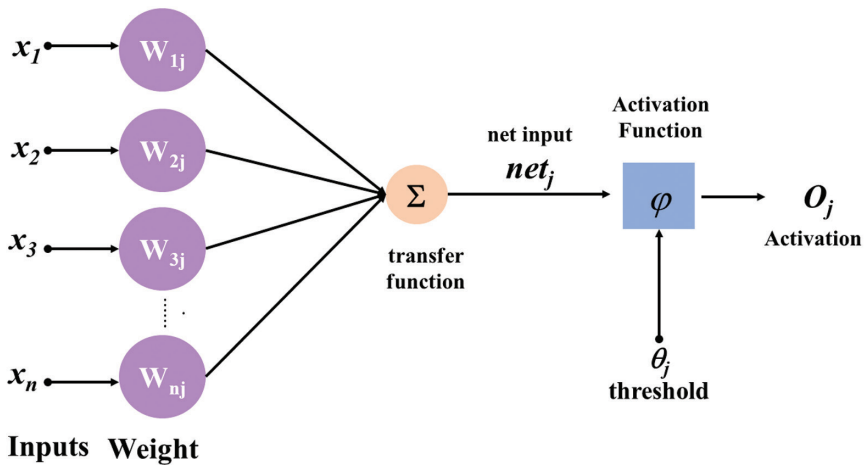


FIGURE 6.5 A neural network.

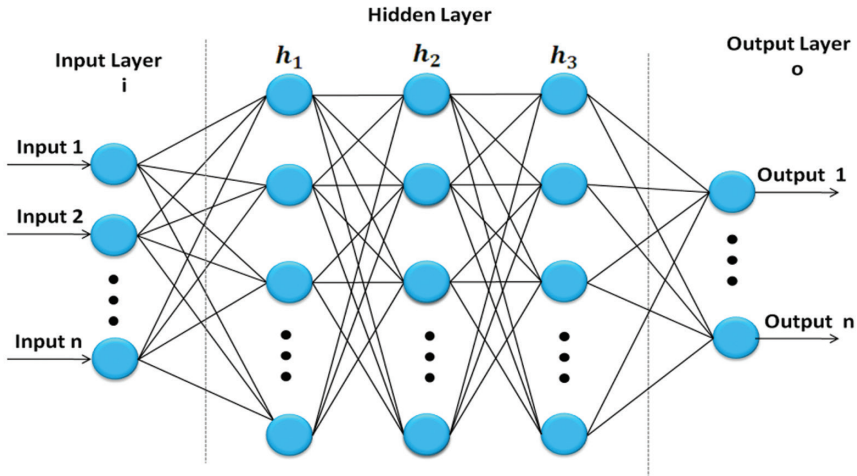


FIGURE 6.6 A deep neural network.

The standard set of real numbers is $[0, 1]$. It is standard practice to use the phrase “fuzzy logic” in a general meaning to refer to a range of conceptual frameworks and methods that concentrate on the systematic management of some kind of degree. This use of the term “fuzzy logic” in a general sense is referred to as “fuzzy logic theory.”

Particularly in engineering settings, the emphasis is placed on finding practical solutions that are flexible enough to accommodate less-than-ideal conditions and lack accuracy (fuzzy control, fuzzy identification, and fuzzy intelligence).

The definition of fuzzy logic that is presented here takes on a more constrained tone. The phrase “fuzzy mathematical logic” has been more common in use in recent years to allude to this idea. In the spirit of conventional, formal logic, it investigates logic through the lens of a plausible representation of incomplete truth.

To describe the Deep Neural Network with Fuzzy Wavelets (DNFW) based on several different fuzzy rules, Takagi, Sugeno, and Kang (TSK) fuzzy frameworks were used. Each fuzzy system was made up of several wavelets transformations, each of which had a programmable interpretation, and the results of these transformations were referred to as dilation variables (Ahmadi et al., 2022).

Each input in a TSK fuzzy model is segmented into fuzzy regions, and a degree of membership in the IF section is allotted to each fuzzy area based on its level of segmentation. In the IF portion of the rules, you will find an implementation of a fixed or linear function of the inputs.

The following are the conditions that must be met while using IF–THEN:

$$P_k : \begin{array}{l} \text{IF } n_1 \text{ is } B_{k1} \dots \text{ AND } n_x \text{ is } B_{kx} \\ \text{THEN } M_k = b_{k0} + b_{k1}n_1 + \dots + b_{kx}n_x \end{array} \quad (6.7)$$